Cyrille E. Mvomo

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Summary

Cyrille (*He/Him* | *Black*) is a clinician-informatician currently pursuing a Ph.D. at the *Human Brain Control of Locomotion Laboratory (HBCL)* at McGill University (Canada) and the *Data Analytics and Rehabilitation Technology Laboratory (DART)* at Lake Lucerne Institute (LLUI, Switzerland). His doctoral research **employs data mining techniques to improve the clinical interpretability of digital mobility biomarkers in Parkinson's disease**. His work has received multiple prestigious distinctions ranging from institutional to international levels, including the **Next Generation Scholar Recognition from IEEE-EMBS, Google, and the US National Science Foundation**. Beyond research, Cyrille's commitment to improving STEM education through innovative teaching and mentoring approaches has also been recognized internationally. He envisions a future as an academic where, on the one hand, he will establish a research program centered on intelligent **wearable systems that generate clinically interpretable insights to inform personalized care in geriatric populations with mobility impairments** and, on the other hand, develop new educational approaches that empower learners with complex needs.

To further discover Cyrille, visit: https://cyrillemvomo.github.io/

[Colored text and icons includes links to proofs, relevant materials or CV sections]

Education

Labels:
Diploma/Certificate
Official Transcripts

Montreal, Canada **McGill University**
2023-Now Ph.D. in Biomechanics & Neurosciences (project R.06 below)
Committee: Dr. Paquette, Dr. Easthope Awai and Dr. Dixon
cGPA: 4.0/4.0 (on going)

Lausanne, CH **Swiss Federal Institute of Technology in Lausanne (EPFL)**
2023 Visiting Student in Bioengineering – Robotics (project R.05 below)
Advisors: Dr. Manzoori and Dr. Bouri
Project GPA: 4.0/4.0

Paris, France **University of Paris – Cité**
2021-2023 M.Sc. in Informatics – Digital Sciences (project R.03/R.04 below)
Concentration in Biomedical Data Science
cGPA: 4.0/4.0 (summa cum laude)

Paris, France **University of Paris – Cité**
2018-2021 B.Sc. in Allied Health *(university title)* (project R.01/R.02 below)
B.Med. in Podiatric Medicine *(clinical title)*
cGPA: 3.7/4.0 (magna cum laude)

Recognitions

24 recognitions for excellence in research, teaching, and service across France, Canada, and Switzerland, spanning institutional to global levels. Secured full PhD funding (CA$120,000), covering living and tuition expenses (Québec resident rate through the France–Québec agreement). Not eligible for major Canadian federal doctoral awards during the first three years of the PhD due to international student status; eligibility restored following a policy change for the 2026–2029 cycle (application under review). Undergraduate and master's studies in France were fully state-funded based on financial need, explaining the absence of major fellowships; funded only to support stays in Switzerland.

Labels:
- *International scope*
- *Federal scope*
- *Provincial scope*
- *Institutional scope*

Awards

2026	Catherine Berg Student Travel Award – CRIR Montreal (CA$500)	*Non-competitive*
2025	Reimagine Education Award (Bronze) – Quacquarelli Symonds (QS) ***[Awarded as one of the six members of the McGill SciLearn team]***	*Competitive*
2025	NextGen Scholar – NSF (USA), IEEE-EMBS and Google (~CA$415)	*Competitive*
2025	Best Paper Award (3rd Place) – IEEE-EMBS BHI'25 (~CA$207) ***[Out of 500+ submissions, top ~0.6%]***	*Competitive*
2025	CRBLM Travel Award – McGill University (CA$700)	*Competitive*
2025	AQSAP Travel Award – Quebec Association for Physical Activity (CA$250)	*Competitive*
2024	ICS Travel Award – Canadian Institute of Health Research (CIHR) (CA$1,700)	*Competitive*
2024	Travel Award – Parkinson Canada (CA$1,000)	*Competitive*
2024	People's Choice Award – CRIR Montreal 3-min Thesis Competition (CA$50)	*Competitive*
2023	Graduated Top 3 M.Sc program (class of 2023) – University of Paris Cité	*Competitive*

Fellowships

2026	Graduate Student Award – Parkinson Canada (CA$40,000)	*Competitive*
2024	Ph.D. Recruitment Fellowship – Quebec BioImaging Network (CA$10,000)	*Competitive*
2023	Graduate Excellence Fellowship – McGill University (CA$60,000)	*Competitive*
2023	Anti-Black Racism Excellence Fellowship – McGill University (CA$10,000)	*Non-competitive*
2023	SMARTS-UP Scholarship – University of Paris Cité (~CA$7,800)	*Competitive*
2022	Visiting Scholarship – Lake Lucerne Institute (~CA$8,700)	*Non-competitive*

Fellowship Under Review

2026	Canada Doctoral Research Scholarship – CIHR (CA$120,000) ***[Selected as one of the 91 best applications forwarded by McGill to CIHR]***	*Competitive*

Honours

2026 Invited contribution to the BHI'25 Special Issue, IEEE Trans. Biomed. Eng. (TBME) Editors
[Selected as one of the best paper at IEEE-EMBS BHI'25]

2025 Reviewer Recognition, IEEE-EMBS BHI'25 Organizing Committee

2025 Invited for Undergraduate Thesis Defense Jury, AFREP Institute, University of Paris – Cité

2025 Guest judge for the 3-minute thesis competition, McGill University

2023 Guest student member of the admission committee, MSc program Digital Sciences U. Paris Cité

2023 Summa Cum Laude, University of Paris – Cité ***[M.Sc. with cGPA>16/20]***

2021 Magna Cum Laude, University of Paris – Cité ***[B.Sc. with cGPA>14/20]***

Grant Preparation

2026 CIHR Project Grant on "Rewiring the brain to alleviate freezing of gait" (CA$1,890,000)
[Contributed to the scientific content of the grant through suggestions for improvement]
PI: Dr. Paquette, Role: Non-applicant, Status: Under Review

Professional Experience

Research

2023 – Now **McGill HBCL & LLUI DART Laboratories**

Montreal & Vitznau Graduate Student Researcher (Project R.06 below)

Canada & CH Referee: Dr. Paquette (McGill Supervisor), Dr. Easthope Awai (LLUI Supervisor)

Responsibilities: Initiated a multi-award-winning Swiss–Canadian collaboration combining neuroimaging, AI, and wearable sensing to improve the clinical interpretability of digital mobility biomarkers in Parkinson's disease. Activities include project conceptualization, funding acquisition, data curation (wearables, neurophysiological, clinical, and patient perception), data analysis, and dissemination to both scientific and community audiences through publications and knowledge translation initiatives. In addition, I mentor undergraduate and master's students and support the conceptualization, data collection, and analysis of fellow PhD students' research projects. I also serve as a reviewer for journals and conferences in digital health and AI, contribute to academic committee activities, and provide support for grant evaluation and writing.

2023 **EPFL Rehabilitation & Assistive Robotics (REHAssist) Group (part of BIOROB & TNE)**

Lausanne, CH Research Intern – Assistive Robotics (Project R.05 below)

Referee: Dr. Manzoori

Responsibilities: Used data mining techniques to evaluate the biomechanical parameters explaining why one mode of active orthotic assistance (e-Walk exoskeleton) is more effective than another in reducing the metabolic cost of walking. Main tasks included multimodal data analysis (inertial motion capture, exoskeleton encoder data, force plates, etc.). Also involved in the evaluation of the active orthosis for mobility support. The role involved collaboratively conceptualizing experiments and conducting multi-terrain data collection (indoor and real-world) of functional outcomes (kinematic, metabolic cost, heart rate, FSR).

2022 **National Institute for Health & Medical Research – U. Paris Cité (INSERM UMR U1284)**

Paris, FR Research Intern – Biomedical Data Science (Project R.04 below)

Referee: Dr. Lhoste

Responsibilities: Prototyped a mobile health application (Flutter SDK) for fall-risk assessment in older adults, based on simple balance tests using mobile sensing and machine learning.

2022 **LLUI DART Laboratory**

Vitznau, CH Research Intern – Neurosciences and Digital Health (Project R.03 below)

Referee: Dr. Easthope Awai

Responsibilities: Collaboratively collected, analyzed, and presented data to clinical partners and international audiences from a project integrating motion capture, virtual reality (VR), and wearable technologies to develop a novel VR-based visual–sensory contribution to balance assessment. The project was part of a larger multicentric pilot trial evaluating the effectiveness of a single session of a technology-assisted fall prevention intervention for older adults. For this larger project, tasks included leading the LLUI arm of the trial, contributing to conceptualization, participant recruitment, data collection (inertial/optical motion capture, force plates, clinical, neurophysiological), preliminary analysis, and collaboration with academic/industry partners.

2021 **UTC Biomechanics & BioEngineering (BMBI) Laboratory** (Project R.01 below)

Compiègne, FR Undergraduate Research Initiation Intern – Biomechanics

Referee: Dr. Ben Mansour

Responsibilities: Extracurricular internship. Conducted a mini-project on the effect of walking speed on peak plantar pressure. Tasks included data acquisition, basic signal processing, interpretation of biomechanical outputs, and results dissemination to lab members.

2021 **Motion Analysis Lab Garches Foundation – Greater Paris University Hospitals (AP-HP)**

Garches, FR Undergraduate Research Initiation Intern – Biomechanics

Referee: Dr. Supiot

Responsibilities: Extracurricular internship. Initiated to motion capture in neurorehabilitation. Assisted researchers during quantified gait analysis sessions with patients.

Clinical

2021 – 2022 **Simone Veil Community Health Center**

Évry, FR Podiatrist Assistant

Referee: Prisca Lonn *(Head Podiatrist)*

Responsibilities: Clinical practice (20%) in podiatric medicine conducted in parallel with the master's degree, with a focus on sports-related conditions, aging populations, and diabetic neuropathy. Activities included anamnesis, mobility assessment (e.g., gait and balance), design of custom medical devices (passive orthoses and prostheses), pharmacological and non-pharmacological treatment prescription, prevention and monitoring of complications, and communication with other healthcare professionals to support coordinated care.

2020 **IMSS Stade Français Paris Rugby Club**

Paris, FR Clinical Training – Sport Podiatric Medicine

Referee: Christophe Blanc *(Head Podiatrist)*

Responsibilities: Clinical training in elite sport (private sector). Assisted a senior sport podiatrist in anamnesis, mobility assessment, design of custom medical devices (passive orthoses), non-pharmacological treatment prescription, prevention and monitoring of complications, and communication with team physicians and therapists (~60 patients).

2018 – 2021 **Greater Paris University Hospitals (AP-HP)**

Paris, FR Clinical Training – General Podiatric Medicine

Referee: Pr. emeritus Matillon *(University of Paris Cité and Lyon 1 University)*

Responsibilities: Clinical training within Europe's largest university hospital center. Weekly rotations across hospital services and a podiatric medicine clinic to provide care to individuals with a wide range of lower-extremity conditions (1,000+ patients). Activities included patient assessment and anamnesis, gait and biomechanical evaluation, review and documentation of medical records, development and implementation of treatment plans (including orthotic management), prevention and monitoring of complications, and interdisciplinary communication with healthcare teams. Hospital services included Internal Medicine, Dermatology, Rheumatology, Endocrinology, Diabetology, Rehabilitation Medicine, and Geriatrics.

Teaching

2024 – Now **Office of Science Education, Faculty of Science, McGill University**

Montreal, CA Science Education Fellow

Referee: Dr. Armin Yazdani *(SciLearn Co-Lead)*

Responsibilities: Member of the McGill SciLearn team. Develop content and teach (in collaboration with cognitive neuroscientists and education developers) the basic principles of neuroeducation to 1000+ undergraduate students and instructors (teaching assistants) in the Faculty of Science. Manage a collaborative learning space dedicated to first-year science undergraduate students. Co-create the "Étudions Ensemble" program, a university-wide initiative aimed at providing McGill students who wish to study in French with a space to study and build connections. Conduct research on the efficacy of neuroeducation interventions on student performance. Organize and facilitate events, including student research showcases and instructor-focused sessions, and meet with McGill administration to promote initiatives and support program development.

2024 – 2025 **Department of Kinesiology and Physical Education, McGill University**

Montreal, CA Teaching Assistant

Referee: Dr. Paquette *(Instructor)*

Responsibilities: Assisted Dr. Paquette in the "Motor Control" course by addressing students' incomprehension, mentoring them during tutorials, contributing to exam design, and supervising and grading both written and oral exams.

2024 – 2025 **Department of Rehabilitation, Faculty of Health, AFREP, University of Paris – Cité**

Paris, FR Lecturer

Referee: Pr. emeritus Matillon *(University of Paris Cité and Lyon 1 University)*

Responsibilities: Developed and delivered 4×20 hours of online courses on the basics of research methodology, with one part in the fall and the other in the winter. Activities included full course design (quizzes, finals, practical, and assignments), as well as coordination of guest lecturers.

Publications

Labels:

* *Equal Contribution, Mentee*
🌎 *International conference*
🏛 *Federal conference*
🏫 *Provincial conference*
🏠 *Institutional conference*

Heavily-reviewed Journal Papers (J)

9 journal publications (1 published, 2 under review, 1 submitted, and 5 in preparation)

J.09 Dahlia Leibovich, **Cyrille E. Mvomo**, Laura Higuita, Justin Wu, Bryana Catedral-Sansalone, Ciara Truong, Ines Nunes Loureiro, Alexandra Potvin-Desrochers, Caroline Paquette. "Freezing of Gait in Parkinson's Disease: Does the New Definition Detect More Freezing?". (2026). *In advanced preparation for Movement Disorders (impact factor 7.6).* ***[Mentee lead author; Preliminary results received 3rd Price Best Undergraduate Oral Presentation at McGill Minds in Motion Conference 2026]***

J.08 Jordan Bedime Songue Njiki, **Cyrille E. Mvomo**, Sara Perfetto, Frédérike Parent-L'Ecuyer, Henri Lajeunesse, Isabella Sierra, Clara Guedes, Jean-Paul Soucy, Alexandra Potvin-Desrochers, Madeleine Sharp, Annie Rochette, Audrey Parent and Caroline Paquette. "Brain Structural and Gait-Related Metabolic Correlates of Fatigue in Parkinson's Disease". (2026). *In advanced preparation for Brain, a Journal of Neurology (impact factor 11.7).*

J.07 **Cyrille E. Mvomo**, Jordan Bedime Songue Njiki, Trina Mitchell, Alexandra Potvin-Desrochers, and Caroline Paquette. "A Multivariate Analysis of Frontoparietal Responses to Complex Walking in Freezing of Gait". (2026). *In advanced preparation for European Journal of Neuroscience (impact factor 2.4).* ***[submission by April 30 for Giladi Special Issue (extended deadline)]***

J.06 **Cyrille E. Mvomo***, Jordan Bedime Songue Njiki*, Dorelle C. Hinton, Tomoha Ogawa, Alexander Thiel, Jean-Paul Soucy, Alexandra Potvin-Desrochers, Caroline Paquette. "The Brain Network that Preserves Gait Adjustments in Parkinson's Disease". (2026). *In advanced preparation for Brain, a Journal of Neurology (impact factor 11.7).*

J.05 **Cyrille E. Mvomo**, Saskia Neumann, Johannes Pohl, Mohamed Elgendi, Chris Easthope Awai. "Beyond Garbage In, Garbage Out: Can Uncertainty Become Insightful in Real-World Outcome Prediction for Parkinson's Disease?". (2026). *In advanced preparation for IEEE Journal of Biomedical and Health Informatics (J-BHI) (impact factor 6.7).*

J.04 Connor Delaney, **Cyrille E. Mvomo**, Valerie Gibson, Chris Easthope Awai, Parag Chitnis, Quentin Sanders. "Assistive Technologies, Clinical Considerations, and Future Directions for Managing Freezing of Gait in Parkinson's Disease". (2026). *Under review in Physical Medicine & Rehabilitation Journal (PM&R) (impact factor 2.8).*

J.03 Audrey Parent, Sara Perfetto, Jordan Bedime Songue Nijki, Frédérike Parent-L'Ecuyer, Isabella Sierra, Henri Lajeunesse, **Cyrille E. Mvomo**, Madeleine Sharp, Annie Rochette, Alexandra Potvin-Desrochers, Caroline Paquette. "The multidimensionality of fatigue in Parkinson's disease: a qualitative study of triggers, experience and coping strategies". (2026). *Submitted to Journal of Neurology (impact factor 4.6).*

J.02 **Cyrille E. Mvomo**, Jordan Bedime Songue Nijki, Dahlia Leibovich, Clara Guedes, Alexandra Potvin-Desrochers, Philippe C. Dixon, Chris Easthope Awai, Caroline Paquette. "Gait-Related Digital Mobility Outcomes in Parkinson's Disease: New Insights into Convergent Validity?". (2026). *Under review in IEEE Transactions on Biomedical Engineering (TBME) (impact factor 4.5).* ***[Preprint] [Invited contribution to BHI'25 Special Issue in IEEE-TBME]***

J.01 Saskia Neumann, **Cyrille E. Mvomo**, Deepak Ravi, Friederike Schulte, Lorenz Assländer*, Chris Easthope Awai*. "Visual Dependence for Postural Control is Increased in Older Adults". (2025). *In Aging and Disease (impact factor 7.0).* [*DOI*]

Heavily-reviewed Conference Papers (C)

1 conference paper (1 published)

C.01 **Cyrille E. Mvomo**, Jordan Bedime Songue Nijki, Sara Perfetto, Dahlia Leibovich, Clara Guedes, Alexandra Potvin-Desrochers, Philippe C. Dixon, Chris Easthope Awai, Caroline Paquette. "Monitoring Parkinson's Disease In-the-Wild". (2025). *In Proceedings of IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI'25)* **[***DOI***]** ***[Acceptance rate <30%, Selected for Oral Presentation, 3rd Place Best Paper Award, 500+ submission, Top 0.6%, NSF IEEE-EMBS Google NextGen, 3 travel awards] [Top venue in Heath Informatics]***

Books Contributions (B)

2 book chapter contribution (1 published and 1 under review)

B.02 **Cyrille E. Mvomo**, Mohamed Elgendi, Chris Easthope Awai. Contribution to "AI and Computational Methods in Rehabilitation". (2026). *Under review for Converging Clinical and Engineering Research on Neurorehabilitation (ICNR 2026)*. ***[Book chapter contribution + Oral if accepted]***

B.01 Cyrille E. Mvomo*, Kira Smith*. "McGill SciLearn Program". (2025). *In Skills For Success 3rd Edition* **[***Proof***]** ***[Oral and Book chapter contribution]***

Conference Abstract (A)

5 conference abstract (2 published, 1 accepted, 2 under review)

A.05 **Cyrille E. Mvomo**, Jean-Christophe Boivin, Hilary Sweatman, Kira Smith, Armin Yazdani, Marcy Slapcoff. "Exploring the Link Between Belief in Neuromyths and Academic Performance in Undergraduate Science Students". *Accepted for SALTISE Conference (2026)*. ***[Abstract + Oral]***

A.04 Jae Young Park, Emilia Morales Zaldivar, Jordan Bedime Songue Nijki, **Cyrille E. Mvomo**, Frederike Parent-L'Ecuyer, Isabella Sierra, Henri Lajeunesse, Audrey Parent, Madeleine Sharp, Annie Rochette, Caroline Paquette. "Examining the Impact of Fatigue on White Matter tracts in Parkinson's Disease". *Under review in Movement Disorders (2026)*. ***[Poster + Abstract if accepted] [Top venue in Parkinson's Disease]***

A.03 Tomoha Ogawa, Clara Guedes, **Cyrille E. Mvomo**, Jordan Bedime Songue Nijki, Leila Hajjaji, Alexandra Potvin-Desrochers, Julia Choi, Caroline Paquette. "Can people with Parkinson's disease improve gait adaptation skills via upregulation of the posterior parietal cortex?". *Under review in Movement Disorders (2026)*. ***[Poster + Abstract if accepted] [Top venue in Parkinson's Disease]***

A.02 Isabella Sierra, Sara Perfetto, Alexandra Potvin-Desrochers, **Cyrille E. Mvomo**, Jordan Bedime Songue Nijki, Henri Lajeunesse, Frederike Parent-L'Ecuyer, Caroline Paquette. "Influence of Fatigue in Parkinson's Disease on Variability of Gait Measures". *In Movement Disorders (2024)* ***[Poster + Abstract] [Top venue in Parkinson's Disease]***

A.01 **Cyrille E. Mvomo**, Jordan Bedime Songue Nijki, Isabella Sierra, Sara Perfetto, Henri Lajeunesse, Alexandra Potvin-Desrochers, Chris Easthope Awai, Caroline Paquette. "Model-Based Gait Outcomes to Track Parkinson's Disease Progression ?" *In Movement Disorders (2024)* ***[Poster + Abstract] [Top venue in Parkinson's Disease]***

Conference Posters (P)

13 conference posters (11 presented, 2 under review)

P.13 Jae Young Park, Emilia Morales Zaldivar, Jordan Bedime Songue Nijki, **Cyrille E. Mvomo**, Frederike Parent-L'Ecuyer, Isabella Sierra, Henri Lajeunesse, Audrey Parent, Madeleine Sharp, Annie Rochette, Caroline Paquette. "Examining the Impact of Fatigue on White Matter tracts in Parkinson's Disease". *Under review for the International Parkinson and Movement Disorder*

Society (MDS) Congress (2026). ***[Poster + Abstract if accepted] [Top venue in Parkinson's Disease]***

- *P.12* Tomoha Ogawa, Clara Guedes, **Cyrille E. Mvomo**, Jordan Bedime Songue Nijki, Leila Hajjaji, Alexandra Potvin-Desrochers, Julia Choi, Caroline Paquette. "Can people with Parkinson's disease improve gait adaptation skills via upregulation of the posterior parietal cortex?". *Under review for the International Parkinson and Movement Disorder Society (MDS) Congress (2026).* ***[Poster + Abstract if accepted] [Top venue in Parkinson's Disease]***
- *P.11* Jordan Bedime Songue Nijki, **Cyrille E. Mvomo**, Sara Perfetto, Isabella Sierra, Henri Lajeunesse, Frédérike Parent-L'Ecuyer, Tomoha Ogawa, Jae Young Park, Madeleine Sharp, Marie-Hélène Pennestri, Alexandra Potvin-Desrochers, Bruno G. G. da Costa, Audrey Parent, Caroline Paquette. "Physical activity and sedentary behavior monitoring in people with Parkinson's disease and fatigue". *(2025). At the International Society of Posture & Gait Research (ISPGR) - World Congress*
- *P.10* Jordan Bedime Songue Nijki, **Cyrille E. Mvomo**, Sara Perfetto, Isabella Sierra, Henri Lajeunesse, Jean-Paul Soucy, Alexandra Potvin-Desrochers, Madeleine Sharp, Annie Rochette, Audrey Parent, Caroline Paquette. "Fatigue in Parkinson's disease is linked to enhanced activity of the fornix and dysregulation of the dorsal attention and central executive networks during complex walking: An [18F]-FDG PET study". *(2024). At the Annual Canadian Movement Disorders Meeting*
- *P.09* **Cyrille E. Mvomo**, Jordan Bedime Songue Nijki, Sara Perfetto, Isabella Sierra, Henri Lajeunesse, Alexandra Potvin Desrochers, Chris Easthope Awai, Caroline Paquette. "Model-Based Gait Outcomes to Track Parkinson's Disease Progression ?" *(2024). At the International Parkinson and Movement Disorder Society (MDS) Congress* ***[2 Travel Awards, Top venue in Parkinson's Disease]***
- *P.08* Isabella Sierra, Sara Perfetto, Alexandra Potvin-Desrochers, **Cyrille E. Mvomo**, Jordan Bedime Songue Nijki, Henri Lajeunesse, Frederike Parent-L'Ecuyer, Caroline Paquette. "Influence of Fatigue in Parkinson's Disease on Variability of Gait Measures". *(2024). At the International Parkinson and Movement Disorder Society (MDS) Congress)* ***[Top venue in Parkinson's Disease]***
- *P.07* Sara Perfetto, Alexandra Potvin-Desrochers, Isabella Sierra, **Cyrille E. Mvomo**, Kevin Shoemaker, Mallar Chakravarty, Caroline Paquette. "Long-term, high-level aerobic training and the age-related decline in cortical thickness". *(2024). At the Society for Neurosciences (SfN) World Congress)* ***[Top venue in Neurosciences]***
- *P.06* Jordan Bedime Songue Nijki, **Cyrille E. Mvomo**, Sara Perfetto, Henri Lajeunesse, Isabella Sierra, Alexandra Potvin-Desrochers, Madeleine Sharp, Rochette Annie, Jean-Paul Soucy, Audrey Parent, Caroline Paquette. "Brain metabolism in Parkinson's disease with and without Fatigue during complex walking: An [18F]-FDG PET study". *(2024). At the Quebec Bioimaging Network (QBIN) Scientific Day)*
- *P.05* **Cyrille E. Mvomo**, Jordan Bedime Songue Nijki, Sara Perfetto, Henri Lajeunesse, Isabella Sierra, Alexandra Potvin-Desrochers, Caroline Paquette. "A gait model that reflects measures of symptoms severity in Parkinson's disease". *(2024). At the 5th Quebec Congress in Adaptation-Rehabilitation Research*
- *P.04* **Cyrille E. Mvomo**, Jordan Bedime Songue Nijki, Sara Perfetto, Henri Lajeunesse, Isabella Sierra, Alexandra Potvin-Desrochers, Caroline Paquette. "A gait model that reflects measures of symptoms severity in Parkinson's disease : A Pilot Study". *(2024). At McGill Minds in Motion Conference*
- *P.03* Jordan Bedime Songue Nijki, **Cyrille E. Mvomo**, Sara Perfetto, Henri Lajeunesse, Isabella Sierra, Alexandra Potvin-Desrochers, Caroline Paquette. "The influence of cognitive load on locomotor adaptation to split-belt treadmill in Parkinson's disease with freezing of gait". *(2024). At McGill Minds in Motion Conference*
- *P.02* Saskia Neumann, **Cyrille E. Mvomo,** Deepak Ravi, Friederike Schulte, Lorenz Assländer, Chris Awai Easthope. "Relative importance of the visual system for balance control is reflected

in the continuous margins of stability while walking". *(2023). At the XXIX Congress of International Society of Biomechanics (ISB)* ***[Top venue in Movement Biomechanics]***

P.01 **Cyrille E. Mvomo**, Deepak Ravi, Friederike Schulte, Lorenz Assländer, Chris Awai Easthope. "Does locomotor perturbation training induce a change in relative sensory contribution to balance? A pilot study". *(2022). At RehabWeek* ***[Top venue in Rehabilitation Engineering]***

Media Coverage

M.04 Funded Research – *Parkinson Canada (2026)*

M.03 SciLearn: Learning how to learn – *The Tribune (2026)*

M.02 KPE PhD Candidate Cyrille Mvomo Honored with IEEE-EMBS BHI 2025 Best Paper Award and NSF-EMBS-Google NextGen Scholar Recognition – *McGill News (2026)*

M.01 Results – CRIR 2024 Student Colloquium – *CRIR News (2024)*

Oral Presentations

Labels:
- *International conference*
- *Federal conference*
- *Provincial conference*
- *Institutional conference*

Conference Presentations (T)

T.04 **Cyrille E. Mvomo**. Epistemic Uncertainty as a Driver of Performance in Wearable Monitoring of Parkinson's Disease: A Real-World Proof-of-Concept Study. *Under review for the International Conference on NeuroRehabilitation (ICNR 2026), Seoul, Korea (Fall 2026) – Technical Audience in AI in Neurorehabilitation.*

T.03 **Cyrille E. Mvomo.** Exploring the Link Between Belief in Neuromyths and Academic Performance in Undergraduate Science Students *at SALTISE, Montreal, Canada (Spring 2026) – Technical Audience in Learning Sciences*

T.02 **Cyrille E. Mvomo.** Monitoring Parkinson's Disease In-the-Wild *at IEEE-EMBS BHI'25, Atlanta, USA (Fall 2025) – Technical Audience in Biomedical Informatics* ***[Selected among Top 9.2% best submissions]***

T.01 **Cyrille E. Mvomo.** McGill SciLearn Program *at Skills for Success Symposium, Montreal, Canada (Fall 2025) – Technical Audience in Learning Sciences*

Invited Conference Presentations

Cyrille E. Mvomo. NextGen Scholar Presentation *at IEEE-EMBS BHI'25, Atlanta, USA (Fall 2025) – Technical Audience in Biomedical Informatics*

Cyrille E. Mvomo. McGill SciLearn Program *at Skills for Success Symposium, Montreal, Canada (Fall 2025) – Technical Audience in Learning Sciences*

Lay Audience Presentations (L)

L.02 **Cyrille E. Mvomo.** What if walking could help track the progression of Parkinson's disease? *TED-like talk at NeuroLingo, Montreal, CA (Fall 2024) – Knowledge Translation* ***[10,000+ views on Youtube]***

L.01 **Cyrille E. Mvomo.** What if walking could help track the progression of Parkinson's disease? *3-minutes elevator pitch at CRIR 2024 Student Colloquium, Montreal, CA (Spring 2024) – Knowledge Translation* ***[People's choice award]***

Guest Lectures (G)

McGill University *Guest Lectures on the Neurosciences of Learning, Memory and Attention*

2025 – 2026 **COMP 202.** Foundations of Programming.

2025 – 2026 **CHEM 222.** Organic Chemistry.

2025 – 2026 **CHEM 110.** General Chemistry 1.

2024 – 2025 **CHEM 120.** General Chemistry 2.

2024 – 2025 **PSYC 204.** Introduction to Psychological Statistics.

Research Projects

R.06 **Cyrille E. Mvomo.** (On going) *What Do We Measure? Advancing the Clinical Interpretability of Digital Mobility Biomarkers in Parkinson's Disease*. McGill Ph.D. Thesis, Montreal, CA

R.05 **Cyrille E. Mvomo.** (2023) *Investigation of Biomechanical Mechanisms Underlying the Efficacy of an Active Hip Exoskeleton Using PCA.* U. of Paris Cité & EPFL Final M.Sc. project, Paris & Lausanne, FR & CH

R.04 **Cyrille E. Mvomo.** (2023) *Identifying Falls-Prone Older Adults Using Smartphone Sensing : A Machine Learning Approach.* U. of Paris Cité & LPI M.Sc. semester project, Paris, FR

R.03 **Cyrille E. Mvomo.** (2022) *Does locomotor perturbation training induce a change in relative sensory contribution to balance? A pilot study.* University of Paris & LLUI M.Sc. semester project, Paris & Vitznau, FR & CH

R.02 **Cyrille E. Mvomo.** (2021) *Effect of foot orthoses on static balance in young adults.* University of Paris B.Sc. Thesis, Paris, FR

R.01 **Cyrille E. Mvomo.** (2021) *Effect of walking speed on peak plantar pressure.* Compiègne University of Technology Internship Thesis, Compiègne, FR

Intellectual Property

2026 **GaitHub**, McGill University
Pre-release of a software to easily analyze movement wearables data.

2025 **PyComplexity**, McGill University
Python module to easily compute the attractor complexity index from wearables.

2023 **MakerRehab**, CRI (LPI), University of Paris - Cité
Prototype of an mHealth technology to predict fall risk from smartphone accelerometers.

2022 **Into the Darkness**, CRI (LPI), University of Paris- Cité
Prototype of a virtual reality experience designed to elicit empathy in people affected by depression.

2022 **Blind Check**, CRI (LPI), University of Paris- Cité
Prototype of a robotic guide dog for people with visual impairments.

Service and Leadership

Social Mobilization

2026 **Founder of Étudions Ensemble**, McGill University
McGill's first university wide program aiming at providing francophone undergraduate students with a space where they can study in French and make connections.

2024 **Patient Engagement Workshop**, Parkinson Canada

Participation in a workshop aimed at improving interaction between researchers, patients, clinicians and industrial partners to enhance patient engagement in Parkinson's research.

2022 **Hackathon**, HackaHealth Zürich
Participated in hackathon to develop solutions for people with disabilities.

2017 **Fundraising Events**, Or Bleu
Participation in soccer tournaments to finance the construction of water wells in Mali.

Organizing Committee

Winter 2026 **SciLearn Orientation Workshop "Prep for Finals"**, McGill University
Co-facilitator and co-organizer of a workshop aimed at providing undergrad science students with basic knowledge of neuroeducation.

Winter 2026 **SciLearn Orientation Workshop "MidTerm Bounce-Back"**, McGill University
Co-facilitator and co-organizer of a workshop aimed at providing undergrad science students with basic knowledge of neuroeducation.

Winter 2026 **Undergraduate Science Showcase 2026**, McGill University
Co-organizer of a conference aimed at providing undergrad science students the opportunity to present their research.

Fall 2025 **SciLearn Orientation Week Workshop**, McGill University
Co-facilitator and co-organizer of a workshop aimed at providing undergrad science students with basic knowledge of neuroeducation.

Fall 2025 **SciLearn TA Training Workshop "How Students Learn"**, McGill University
Co-facilitator and co-organizer of a workshop aimed at providing teaching assistants in Chemistry with basic knowledge of neuroeducation.

Winter 2025 **SciLearn Orientation Workshop "Prep for Finals"**, McGill University
Co-facilitator and co-organizer of a workshop aimed at providing undergrad science students with basic knowledge of neuroeducation.

Winter 2025 **SciLearn Orientation Workshop "MidTerm Bounce-Back"**, McGill University
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Fall 2024 **SciLearn Orientation Workshop "MidTerm Bounce-Back"**, McGill University
Co-facilitator and co-organizer of a workshop aimed at providing undergrad science students with basic knowledge of neuroeducation.

Winter 2023 **AIRE Master Seminar 2023 "AI & Medicine"**, CRI (LPI), University of Paris
Co-organizer of a workshop on AI in medicine.

Peer Reviewing

2025 **IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI'25)**
Ad hoc review of 6 articles on Precision Health and AI. ***[Reviewer Recognition]***

2024 **Medicine & Science in Sports & Exercise**
Ad hoc review (via supervisor) of an article on wearable technologies for monitoring mobility disorders in people with Parkinson's disease.

Teaching

McGill University

Fall 2025	**Teaching Assistant –** EDKP 447 (Motor Control).	~70 Students
Fall 2024	**Teaching Assistant –** EDKP 447 (Motor Control).	~60 Students

University of Paris – Cité (AFREP)

Fall 2025	**Graduate Student Instructor –** UE 5.4 (Research Initiation).	~25 Students
Winter 2025	**Graduate Student Instructor –** UE 5.3 (Research Initiation).	~25 Students
Fall 2024	**Graduate Student Instructor –** UE 5.4 (Research Initiation).	~30 Students
Winter 2024	**Graduate Student Instructor –** UE 5.3 (Research Initiation).	~30 Students

Mentoring

Graduate Students

2026-Now ***Florian Gourdin**, Visiting M.Eng. Student, Polytech Marseille* *Biomedical Engineering*
Project: Agreement between inertial and optical measures of step length during treadmill walking in Parkinson's Disease

2024-Now ***Jae Young Park**, M.Sc. Student, McGill University* *Biomechanics & Neuroscience*
Project: Fatigue-Related White Matter Changes in Parkinson's Disease

2024-Now ***Tomoha Ogawa**, M.Sc. Student, McGill University* *Biomechanics & Neuroscience*
Project: Neuromodulation of the parietal cortex to improve gait in Parkinson's Disease

2024-Now ***Clara Guedes**, M.Sc. Student, McGill University* *Biomechanics & Neuroscience*
[2026 Canada CIHR Master Research Scholarship Recipient]
Project: Split-belt treadmill training to improve gait in Parkinson's Disease

Undergraduate Students

2026-Now ***Justin Wu**, B.Sc. Student, McGill University* *Kinesiology*
Project: Annotating Freezing of Gait Recordings in Parkinson's Disease

2026-Now ***Ines Nunes Loureiro**, B.Sc. Student, McGill University* *Kinesiology*
Project: Annotating Freezing of Gait Recordings in Parkinson's Disease

2026-Now ***Bryana Catedral-Sansalone**, B.Sc. Student, McGill University* *Kinesiology*
Project: Annotating Freezing of Gait Recordings in Parkinson's Disease

2026-Now ***Ciara Truong**, B.Sc. Student, McGill University* *Kinesiology*
Project: Annotating Freezing of Gait Recordings in Parkinson's Disease

2024-2026 ***Dahlia Leibovich**, B.Sc. Student, McGill University* *Kinesiology*
Project: Evaluating Freezing Severity in Parkinson's Disease
Next: M.Sc. Student in Kinesiology, University of Toronto

2024-2025 ***Clara Guedes**, B.Sc. Student, McGill University* *Kinesiology*
Project: Impact of Sex on Brain Function during Gait in Parkinson's Disease with Fatigue
Next: M.Sc. Student in Biomechanics & Neuroscience, McGill University

2024 ***Laura Higuita**, Visiting B.Sc. Student, UAO Columbia* *Biomedical Engineering*
[Mitacs Globalink Summer Intern Program]

Project: Detecting Freezing of Gait in Parkinson's Disease

Next: Bioengineer

Relevant Training

2024-Now **McGill Science Education Fellow Program**, McGill University
Comprehensive training as instructor

2024 **Researchers and Entrepreneurs Network Program (RCE)**, V1 Studio
Entrepreneurship training to translate research into real-world impact

2023 **Integrating sex and gender into health research**, Canadian Institute of Health Research
Training on fairness in research

2023 **TCPS 2 CORE-2022 (Research Ethics)**, Government of Canada
Training on ethics in research

2022 **Gait Analysis & Rehabilitation using CAREN (Operator Level 1)**, Motek Medical
Training on fully immersive motion analysis

2021 **AFGSU 1/2 (emergency care)**, French Ministry of Health

Professional Affiliations & Licensure

2025-Now **Student Member**, Institute of Electrical and Electronics Engineers (IEEE)

2025-Now **Student Member**, IEEE Engineering in Medicine and Biology Society (EMBS)

2024-Now **Student Member**, Movement Disorders Society (MDS)

2023-Now **Student Member**, Centre for Interdisciplinary Research in Rehabilitation (CRIR) Montreal

2024-Now **Student Member of the Committee on Innovative Practices in Research Ethics**, CRIR

2023-Now **Student Member**, CRBLM, Faculty of Medicine and Health Sciences, McGill University

2023-2026 **Student Member**, Quebec BioImaging Network (QBIN)

2023-Now **Member of the Research Committee, Rehabilitation Department**, University of Paris – Cité

2021-2023 **Member,** French Podiatric Medicine Society (SOFPOD)

2021-2023 **Registered Podiatrist,** French National Council of Podiatrists (ONPP)

References

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